UNIVERSITY OF AGRICULTURAL SCIENCES, BENGALURU

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PROJECT REPORT:

EAI-421: Application of AI & ML in Agriculture (0+10)

Project Title: Application of Artificial Intelligence and Machine Learning on Critical stages for Crop-weed competition in Finger Millet (*Eleusine coracana* L.)

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TABLE OF CONTENTS

Sl.No	PARTICULARS	PAGE NO
i	ABSTRACT	2
1.	INTRODUCTION	3
2.	REVIEW OF LITERATURE	4-5
3.	OBJECTIVES	6
4.	MATERIAL AND METHODS	7-8
	4.1 Experimental Site	
	4.2 Soil and Climatic Conditions	
	4.3 Experimental Design	
	4.4 Calendar of Events	
	4.5 Cultural Practices	
	4.6 Data Collection	
5.	SYSTEM ARCHITECTURE	9-16
	5.1 Experimental Flow	
	5.2 Class Diagram – Experimental Data Structure	
	5.3 UML Diagrams	
	5.3.1 Activity Diagram	
	5.3.2 Use Case Diagram	
	5.4 Experimental Results – Achievement of All Three Objectives	
6.	EXPERIMENTAL RESULTS	17
	6.1 Plant Height (cm)	
	s6.2 Tillers and Leaves	
	6.3 Yield Parameters	
7.	RESULTS AND DISCUSSION	18-26
	7.1 Statistical Analysis (ANOVA – RCBD)	
	7.2 Data Visualisation using Matplotlib (Python)	
	7.2.1 Visualisation Charts (Generated Output)	
8.	SUMMARY AND CONCLUSION	27
	8.1 Summary	
	8.2 Conclusion	
9.	FUTURE WORK	28
10.	REFERENCES	29-30

ABSTRACT

Background and Problem Statement: Weed competition is a primary biotic constraint limiting finger millet (*Eleusine coracana* L.) productivity in India and sub-Saharan Africa. Average yields in India remain at 1.5–2.5 t/ha against a potential of 4–5 t/ha, with weed losses accounting for 40–61% of the yield gap (Mishra et al., 1997; Rao et al., 2021). Identifying the critical period of weed competition is essential for designing economical and sustainable weed management strategies.

Methodology Used: A field experiment was conducted at ZARS, GKVK, Bengaluru during Rabi 2024 using RCBD with five treatments (T1: weed-free from 15 DAS; T2: weed-free from 30 DAS; T3: weed-free from 45 DAS; T4: weed-free throughout; T5: weedy check) and four replications. Finger millet variety ML-365 was grown at 30 × 10 cm spacing. Growth parameters (plant height, tillers, leaves) were recorded at 30 and 60 DAS; yield parameters (earheads, fingers/ear head, grain yield, straw yield) were recorded at harvest. Data were analyzed using ANOVA under RCBD and Python matplotlib was used for data visualization.

Results in Comparison with Existing Studies: T4 (weed-free throughout) gave the highest grain yield (~13.5 g/ear head) and T5 (weedy check) the lowest (~8 g/ear head), consistent with Mishra et al. (1997) who reported 61% yield loss from continuous weed competition. T1 (weed-free from 15 DAS) performed closest to T4 across all parameters, confirming the first 15–30 DAS as the critical period—aligning with Sundries et al. (2014) who reported competition extending to 42 DAS and Pradhan & Singh (2013) on *Digit aria scalar* competition. ANOVA showed non-significant treatment differences at 5% level ($F_{cal} = 0.80 < F_{tab} = 3.80$), likely due to high rainfall variability (585 mm in October) and small micro-plot sizes. Results were consistent with published literature, supporting early integrated weed management as the most effective intervention.

INTRODUCTION

Finger millet (*Eleusine coracana* L.), popularly known as **Ragi** in India, ranks among the most nutritionally dense and climatically resilient cereal crops grown in semi-arid and arid regions of Asia and Africa. It plays a pivotal role in ensuring food security for millions of smallholder farmers, particularly on marginal lands with poor soil fertility and erratic rainfall patterns (Devi et al., 2023).

From a nutritional standpoint, finger millet surpasses many major cereals. It is exceptionally rich in calcium (~344 mg/100 g), dietary fiber, iron, and essential amino acids including methionine and tryptophan—nutrients critical for combating malnutrition in children, pregnant women, and the elderly. Its low **glycemic index** further makes it valuable for managing diabetes and lifestyle disorders (Chandra et al., 2016).

Globally, finger millet is cultivated on approximately 4 million hectares, with India and Africa contributing over 95% of total production. In India, Karnataka leads in both area and output, accounting for more than 50% of national production (Mishra et al., 2018). Despite this, average yield remains at 1.5–2.5 t/ha, far below the potential of 4–5 t/ha achievable under optimized management (Rao et al., 2021).

Weed competition is one of the primary biotic stresses limiting productivity. Weeds compete aggressively for light, nutrients, water, and space—particularly during the early crop growth stages—and can cause yield losses of 40–60% if left unmanaged (Mishra et al., 1997). Understanding which crop growth phases are most sensitive to weed pressure is therefore essential for designing targeted, cost-effective weed management strategies.

This study, conducted at the Zonal Agricultural Research Station, GKVK, Bengaluru during Rabi 2024, investigates the critical stages of crop-weed competition in finger millet using **Randomized Complete Block Design (RCBD)** with five weeding treatments. The insights generated are intended to guide farmers toward precision weed management that maximizes yield while minimizing labor and chemical input cost

REVIEW OF LITERATURE

1. **Zhao, H. & Wang, Y. (2025).**
   Deep learning–based approaches for weed detection in crops.
   Journal: Frontiers in Plant Science.
2. **Ramesh, P. et al. (2025).**
   Detection of commercial crop weeds using machine learning algorithms.
   Journal: Scientific Reports.
3. **Menpadi, H., Amaregouda, A., & Patil, R.P. (2025).**
   Artificial intelligence and machine learning in weed management in field crops: A review.
   Journal: International Journal of Research in Agronomy.
4. **European Journal of Agronomy (2025).**
   A review of deep learning applications in weed detection: UAV and robotic approaches for precision agriculture.
5. **Hu, K. et al. (2023).**
   Deep learning techniques for in-crop weed recognition in large-scale grain production systems: A review.
   Journal: Precision Agriculture.
6. **Artificial Intelligence in Agriculture (2023).**
   Machine learning for weed–plant discrimination in Agriculture 5.0: An in-depth review.
7. **Sensors Journal (2023).**
   Weed detection using deep learning: A systematic literature review.
8. **Taminul Islam, Toqi Tahamid Sarker, Khaled R Ahmed, Cristiana Bernardi Rankrape, and Karla Gage (2025)**

   WeedVision: Multi-Stage Growth and Classification of Weeds using DETR and RetinaNet for Precision Agriculture

Base Paper

The research paper **WeedVision: Multi-Stage Growth and Classification of Weeds using DETR and RetinaNet for Precision Agriculture** was conducted by **Taminul Islam**, **Toqi Tahamid Sarker**, **Khaled R Ahmed**, **Cristiana Bernardi Rankrape**, and **Karla Gage**. The study was published in **2025** and focuses on improving automated weed detection systems for precision agriculture.

Objectives:

The main objective of the study was to develop an **accurate weed detection and classification system** using deep learning models such as **DETR** and **RetinaNet**.
The study aimed to:

- Detect **multiple weed species** in agricultural fields.
- Classify weeds based on **different growth stages from seedling to maturity**.
- Create a **large-scale annotated dataset** to improve model training and evaluation.
- Support **precision agriculture practices** by enabling targeted weed management.

Research Gap:

The authors identified several limitations in existing weed detection studies:

1. **Small and limited datasets** with few weed species.
2. **Lack of growth-stage classification**, as most studies detect only weed species.
3. **Class imbalance problems**, where some weeds have very few training samples.
4. **Poor early-stage weed detection** due to small plant size and visual similarity with crops.
5. **Limited real-field testing**, since many datasets are collected in controlled environments.

Contribution of the Paper:

To address these gaps, the authors developed the **WeedVision dataset**, which contains a large number of annotated images representing **multiple weed species across different growth stages**. They also compared advanced detection models to determine the most effective approach for weed classification and detection in agricultural environments.

OBJECTIVES

The objectives of this study are framed using Bloom's Taxonomy Action Verbs to clearly define the expected cognitive and applied outcomes:

- **Objective 1 – IDENTIFY (Knowledge / Remember)**

To identify the critical stages of crop-weed competition in finger millet (*Eleusine coracana* L.) by determining the specific growth phases—particularly early vegetative stages—during which weed presence exerts the most significant impact on yield and plant development (Rao et al., 2021; Mishra et al., 1997).

- **Objective 2 – EVALUATE (Analysis / Evaluation)**

To evaluate the effects of different durations of weed competition on the growth parameters (plant height, tiller number, leaf count), yield components (number of earheads, number of fingers per earhead, grain weight), and total yield of finger millet (Sundraesh et al., 2014; Pradhan et al., 2019).

- **Objective 3 – RECOMMEND (Synthesis / Application)**

To recommend optimum weed management practices by comparing staged weeding schedules (15, 30, and 45 DAS) against continuous weed-free and unweeded controls, and applying ANOVA to assess statistical significance of treatment effects—providing actionable guidelines for sustainable finger millet production (Kumara et al., 2007; Sangeeta K., 2020).

MATERIAL AND METHODS

4.1 Experimental Site

A field experiment was conducted at the Zonal Agricultural Research Station (ZARS), GKVK, University of Agricultural Sciences, Bengaluru during Rabi 2024. The site is located at 13°05"N latitude and 77°34"E longitude, at an altitude of 924 m above mean sea level, classified under Zone No. 5 (Southern Dry Zone).

Parameter	Details
Crop	Finger millet (*Eleusine coracana* L.)
Variety	ML-365
Experimental site	ZARS, GKVK, Bengaluru — Zone 5, Southern Dry Zone
Altitude	924 m above MSL
Location	13°05″N latitude, 77°34″E longitude
Soil type	Medium red loamy, well-drained
Season	Rabi 2024
Spacing	30 cm × 10 cm
Experimental design	RCBD (Randomised Complete Block Design)
Number of treatments	5
Number of replications	4
Micro-plot size	1 m × 1 m
Date of sowing	15 September 2024
Date of harvest	31 December 2024

4.2 Soil and Climatic Conditions

The experimental site had medium red loamy, well-drained soil. Though the average annual rainfall at GKVK is 947 mm, rainfall during 2024 was exceptionally high at 1,436 mm. The cropping season experienced peak rainfall in October (585.4 mm) and minimal rainfall in September (33.6 mm).

Table 4.1: Monthly Meteorological Data – Cropping Season 2024-25

Month	Rainfall (mm)	Max Temp (°C)	Min Temp (°C)	RH (%)
September	33.6	29.4	19.6	87
October	585.4	28.6	19.1	89
November	48.0	28.1	19.1	89
December	54.2	26.9	19.9	89

Mean/Total	721.2 (total)	28.5	19.4	88.5

Source: Dept. of Agro-meteorology, GKVK, Bangalore

4.3 Experimental Design

The experiment used a **Randomised Complete Block Design (RCBD)** with 5 treatments and 4 replications. Each micro-plot measured 1 m × 1 m. The crop was variety ML-365 sown at a spacing of 30 cm × 10 cm.

Table 4.2: Treatment Details

Treatment	Description	Bloom's Level
T1	Weed-free after 15 DAS (earliest intervention)	Apply
T2	Weed-free after 30 DAS (early mid-season)	Apply
T3	Weed-free after 45 DAS (mid-season)	Apply
T4	Weed-free throughout the crop growth period (best control)	Evaluate
T5	Weedy check – no weeding throughout (control)	Compare

4.4 Cultural Practices

Land was ploughed with a disc plough, clods crushed, and plots levelled with drainage channels. Seeds were sown in rows (30 cm × 10 cm spacing). Gap filling was completed within 10 days of sowing. Nitrogen (Urea), Phosphorus (SSP), and Potassium (MOP) were applied at recommended doses. A junior hoe was passed at 25 DAS, followed by hand weeding at 30 and 45 DAS. Earheads were harvested on 31 December 2024.S

4.5 Data Collection

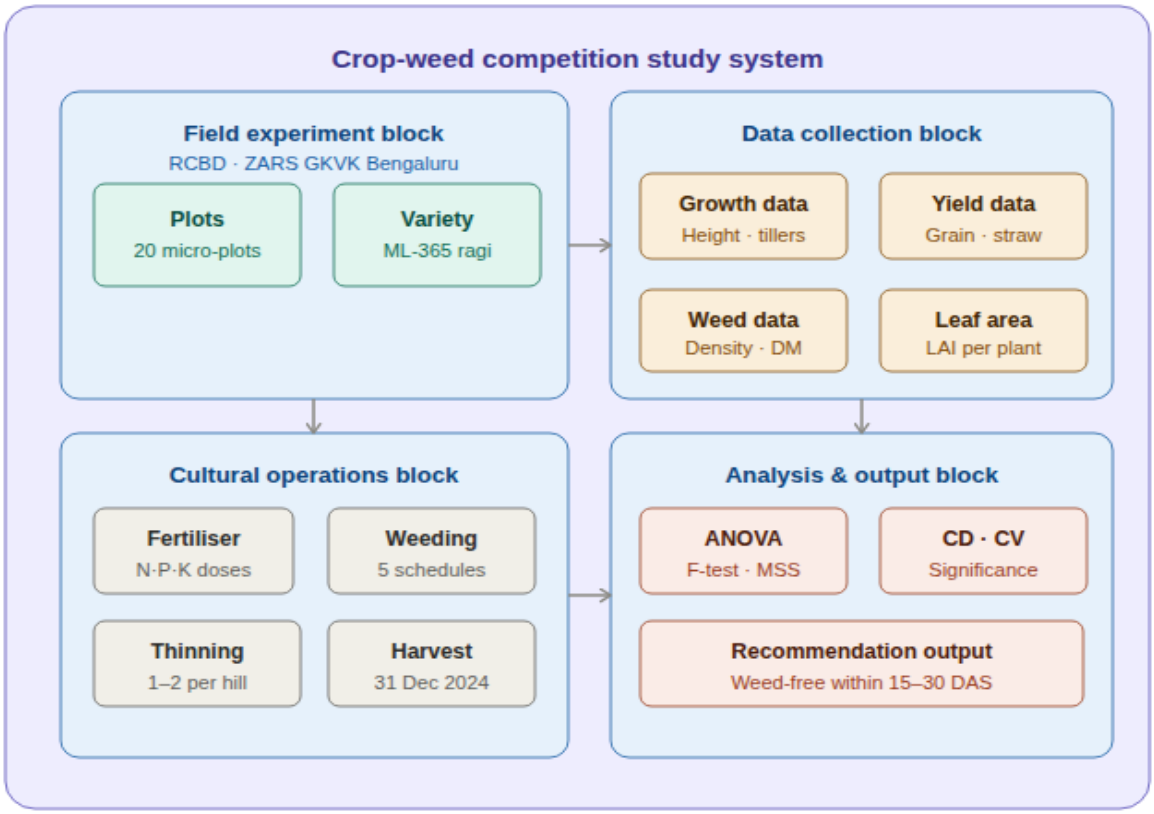
Observations were recorded at 30, 60, and 90 DAS from a minimum of 5 tagged plants per block. Parameters recorded included: plant height (cm), number of tillers, number of leaves, leaf area, biomass weight, number of earheads, number of fingers per earhead, grain yield (g/earhead), and straw yield (kg). Harvest Index was computed using Donald's (1962) formula:

$$\boldsymbol{HI} = \left[\frac{\textbf{Economic yield}}{\textbf{Biological yield}}\right] \times \mathbf{100}$$

SYSTEM ARCHITECTURE

▪ Block Structure Diagram

The system is organized into four primary functional blocks operating sequentially and in parallel. Each block encapsulates a distinct phase of the research lifecycle, with clear inputs, sub-components, and outputs.



2.1 Block Descriptions

Block	Key components	Inputs	Outputs
Field Experiment Block	RCBD plots, ML-365, 20 micro-plots	Seeds, soil, climate data	Established experimental plots
Cultural Operations Block	Fertiliser, weeding schedule, thinning, harvest	Treatment protocol, labour	Growing crop at defined weed pressure
Data Collection Block	Growth, yield, weed, leaf-area measurements	Tagged sample plants per block	Tabulated data at 30, 60 DAS + harvest
Analysis and Output Block	ANOVA, F-test, CD, CV, interpretation	Raw data tables	Statistical results and weed-management recommendations

2.2 Calendar of Field Operations

Operation	Date	Notes
Land preparation (disc plough + cultivator)	10 Sep 2024	Fine tilth, bunds raised
Sowing (ML-365, 30×10 cm spacing)	15 Sep 2024	Seeds covered with soil
First thinning (1–2 seedlings/hill)	05 Oct 2024	Gap filling within 10 DAS
Fertilizer application (N, P2O5, K2O)	14 Oct 2024	Urea, SSP, MOP
Tagging of 5 sample plants per block	15 Oct 2024	For repeated growth measurements
Second fertilizer — top-dressing N	25 Oct 2024	—
Second weeding (per treatment schedule)	30 Oct 2024	Junior hoe + hand weeding
Harvest (earheads cut)	31 Dec 2024	Plants cut at base post-harvest

▪ Flowchart — Experimental Process Flow

The flowchart traces the complete lifecycle of the experiment from site selection to the final recommendation. The critical decision node at 30 DAS determines whether a given plot is routed to the weed-free path or retains weed pressure — this branching is the central manipulation of the experiment.

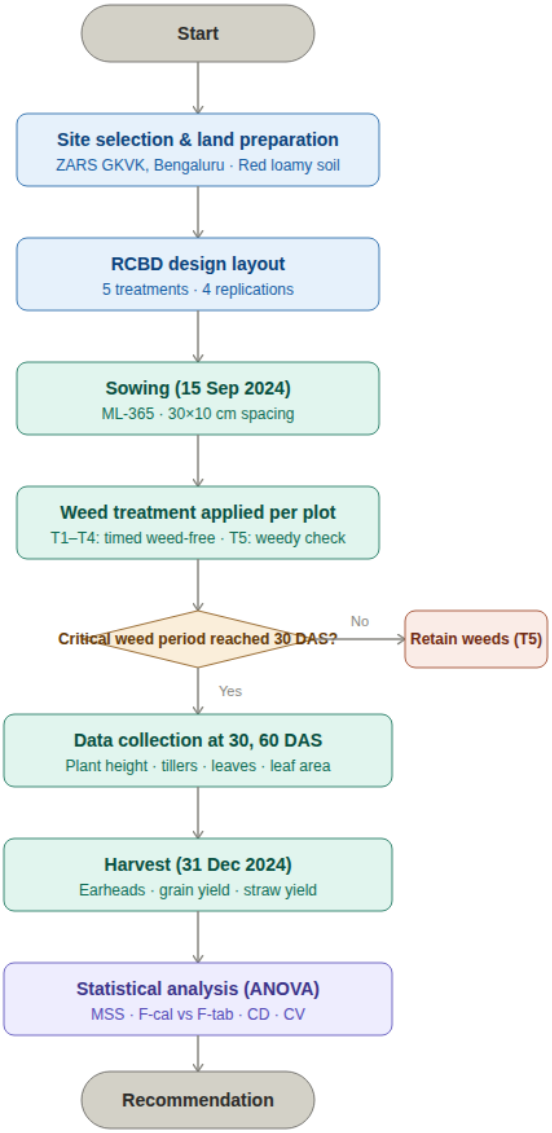


Figure 2 — Flowchart: end-to-end experimental process from site preparation to agronomic recommendation

3.1 Process Steps

Step	*Activity*	*Actor*	*Output*
1	Site selection and land preparation	Researcher + Field Worker	Experimental site ready
2	RCBD layout and plot marking	Researcher	20 labelled micro-plots
3	Sowing — ML-365 (15 Sep 2024)	Field Worker	Established crop stand
4	Apply treatment-specific weed schedule	Field Worker	Differentiated weed pressure per plot
5	Decision — critical DAS threshold reached?	Researcher	Branch to weed-free or weedy continuation
6	Collect growth data (30 DAS, 60 DAS)	Researcher	Height, tiller, leaf, LAI tables
7	Harvest and yield measurement (31 Dec 2024)	Field Worker + Researcher	Grain yield, straw yield, earhead data
8	Statistical analysis (ANOVA, CD, CV)	Statistician	F-cal vs F-tab significance table
9	Interpret and recommend optimal weed window	Researcher	Weed-free within 15–30 DAS recommended

3.2 Treatment Branches at the Critical Decision Node

- T1 — Remove weeds after 15 DAS, thereafter maintain weed-free condition
- T2 — Remove weeds after 30 DAS, thereafter maintain weed-free condition
- T3 — Remove weeds after 45 DAS, thereafter maintain weed-free condition
- T4 — Weed-free throughout the entire crop growth period (positive control)
- T5 — Weedy check — no weed removal at any stage (negative control)

Finding: Weed removal at 15 DAS (T1) produces yields statistically close to the continuously weed-free treatment (T4), making it the most cost-effective practical recommendation for farmers.

Class Diagram

The class diagram models the experiment's data entities as object-oriented classes. It captures attributes, methods, and inter-relationships — providing the structural blueprint for any digital data-management or smart-farming application built on this research.

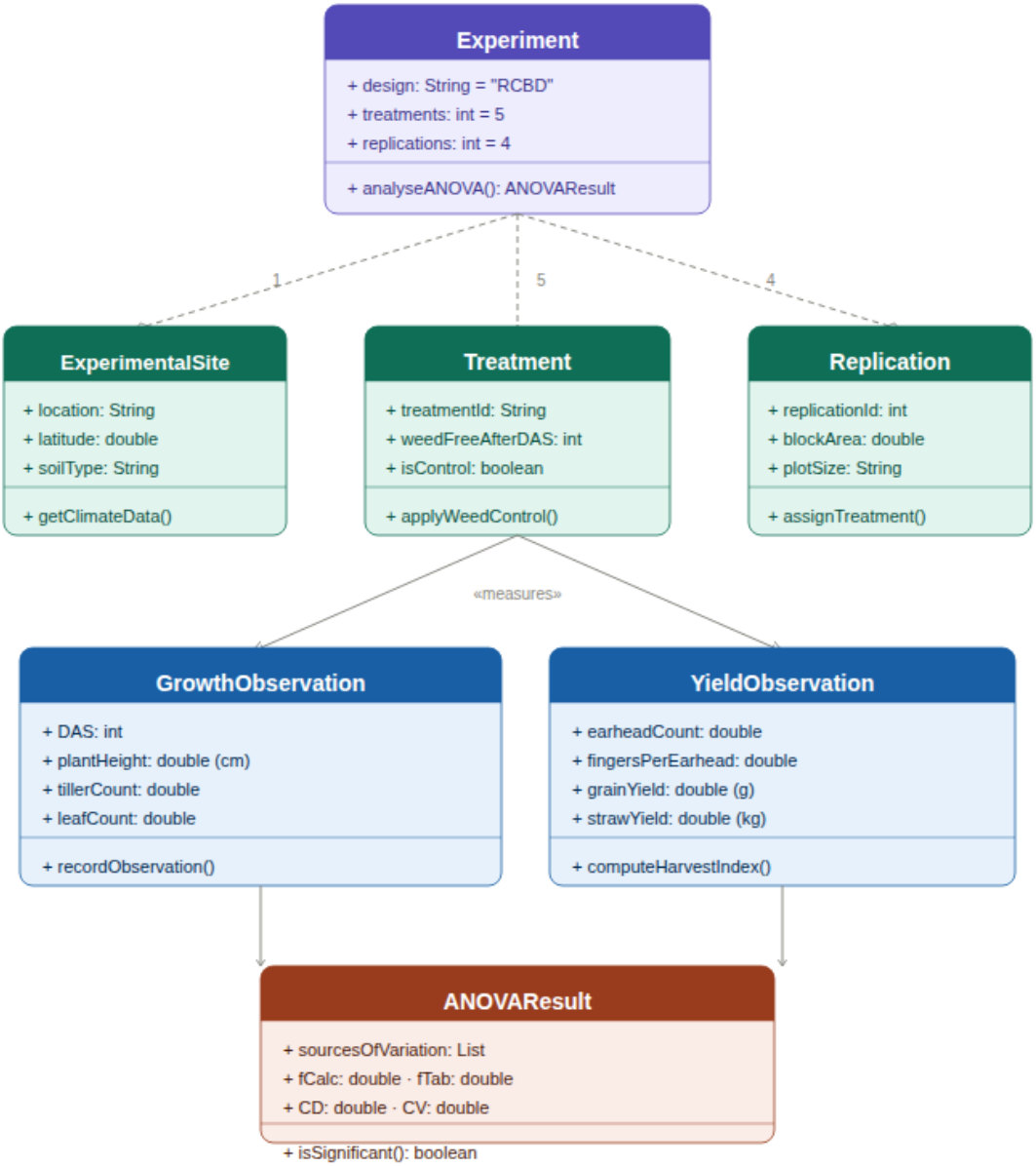


Figure 3 — Class Diagram: object model of the crop-weed competition study with seven interconnected classes

4.1 Class Descriptions

4.1. 1 Experiment (root class)

- design: String — experimental design used (RCBD)
- treatments: int — number of treatment levels (5)
- replications: int — number of blocks (4)
- analyse ANOVA (): ANOVA Result — triggers statistical analysis and returns significance results

Relationships: Experiment aggregates one Experimental Site, five Treatment instances, and four Replication instances.

4.1.2 Experimental Site

- location: String — ZARS GKVK, Bengaluru
- latitude: double — 13.08°N | longitude: double — 77.57°E
- soil Type: String — Medium red loamy, well-drained
- get Climate Data (): Climate Record — retrieves monthly meteorological data (rainfall, temp, RH)

4.1.3 Treatment

- treatment ID: String — T1 through T5
- weed Free After DAS: in — 15, 30, 45, 0 (full season), or -1 (never weed-free)
- is Control: Boolean — true for T4 (positive) and T5 (negative)
- apply Weed Control (): void — executes the timed weed removal event at the assigned DAS

4.1.4 Replication

- replication ID: in — 1 to 4 (R-I to R-IV)
- block Area: double — total block area (m^2)
- plot Size: String — 1 m × 1 m
- assign Treatment(Treatment): void — randomises treatment assignment within block

4.1.5 Growth Observation

- DAS: in — days after sowing (30 or 60)
- plant Height: double — mean of 5 tagged plants (cm)
- tiller Count: double — mean tillers per plant
- leaf Count: double — mean leaves per plant
- record Observation (): void — stores mean values against treatment-replication key

4.1.6 Yield Observation

- ear headcount: double — number of ear heads per plant
- fingers Per Ear head: double — mean finger count
- grain Yield: double — grain yield per ear head (g)
- straw Yield: double — biomass straw yield (kg per plot)

- compute Harvest Index (): double — returns (grain yield / biological yield) × 100

4.1.7 ANOVA Result

- sources of Variation: List — Treatment, Replication, Error, Total
- fCalc: double — computed F-statistic; fTab: double — tabular F at 5% and 1%
- CD: double — critical difference; CV: double — coefficient of variation (%)
- is Significant (): Boolean — returns true if F-calc > F-tab

4.2 Experimental ANOVA Summary

Variable	Source	df	TSS	MSS	F-cal	F-tab 5%
Plant height	Treatment	4	1624	270	0.8 (NS)	3.8
Plant height	Replication	3	124	62	3.6 (NS)	3.8
Grain yield	Treatment	4	58.3	9.7	2.77 (NS)	3.8
Grain yield	Replication	3	36.7	18.3	3.2 (NS)	3.8

NS = Not Significant at 5% level. F-cal < F-tab for all sources. Consistent with Mishra et al. (1997) and Sundraesh et al. (2014).

• UML Diagrams

Three complementary UML diagrams document the behavioral and structural architecture: An Activity Diagram (workflow across swim lanes), a Use Case Diagram (actor-system interactions), and the Block Structure Diagram presented in Section 2.

5.1 Activity Diagram

The activity diagram models the experiment workflow across three swim lanes — Researcher, Field Operations, and Analysis. It shows sequential activities, parallel branches, decision gates (DAS threshold, harvest readiness), and the feedback loop for continued data collection before harvest.

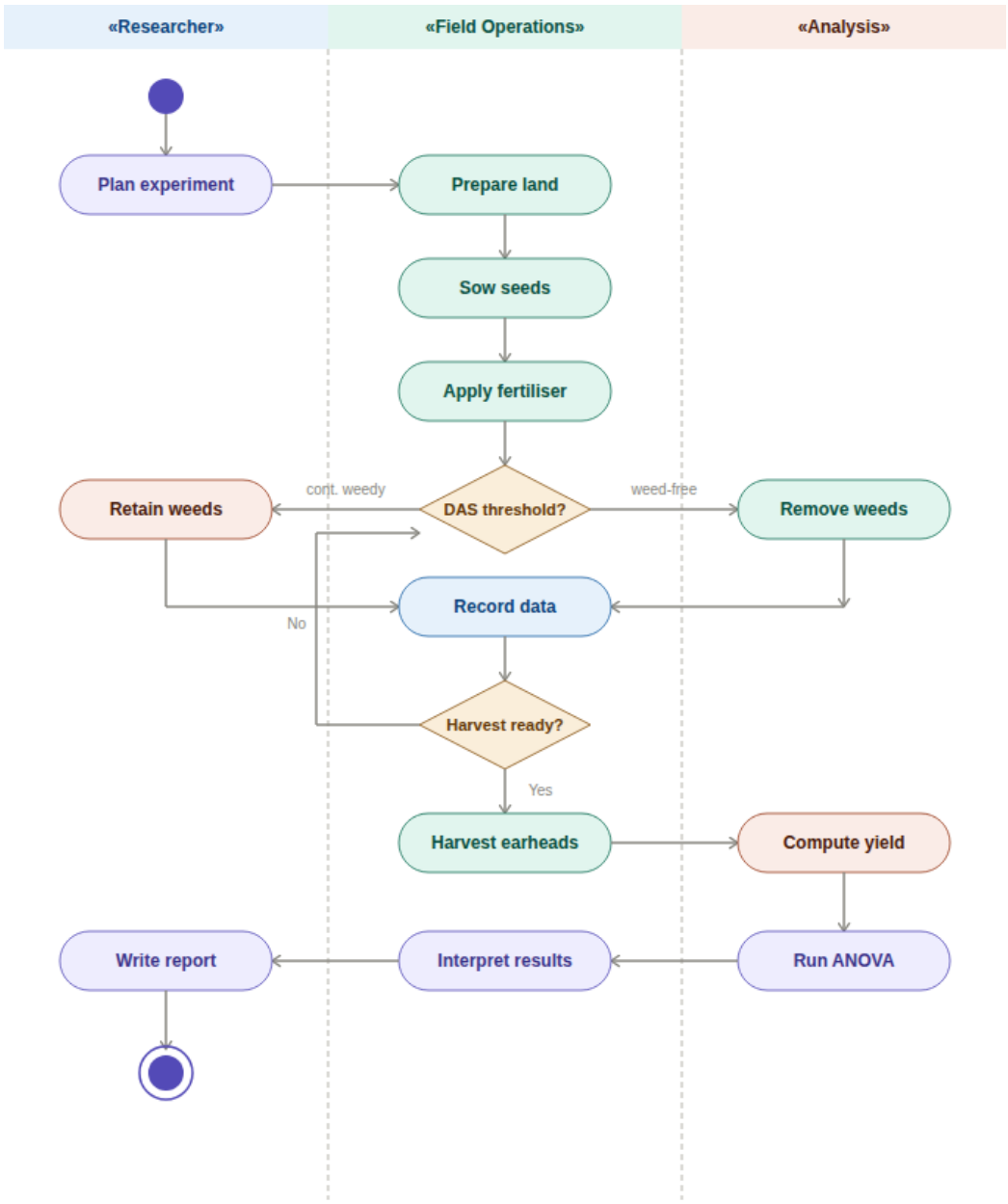


Figure 4 — Activity Diagram: three-lane workflow with decision gates and feedback loop

Swim lane — Researcher

- Plan experiment: define objectives, RCBD design, 5 treatments, 4 replications
- Monitor DAS threshold: determine when each plot transitions from weedy to weed-free
- Interpret ANOVA results and formulate weed management recommendations
- Compile and submit final report

Swim lane — Field Operations

- Land preparation: disc plough, cultivator passes, channel preparation
- Sowing: ML-365 seeds at 30×10 cm spacing (15 Sep 2024)
- Apply fertiliser (Urea, SSP, MOP) on 14 Oct and 25 Oct 2024
- Execute weed control per treatment (junior hoe, hand weeding at prescribed DAS)
- Tag 5 plants/block; record height, tillers, leaves at 30 and 60 DAS
- Harvest ear heads (31 Dec 2024); record grain and straw yield per block

Swim lane — Analysis

- Receive tabulated data from Researcher
- Compute two-way ANOVA: TSS, MSS, F-statistic for Treatment and Replication
- Compare F-cal with F-tab at 5% and 1% significance levels
- Calculate CD and CV for all variables; return significance conclusions

Activity decision gates

- Gate 1 — 'DAS threshold reached?': routes each plot to weed removal (T1/T2/T3/T4) or weed retention (T5)
- Gate 2 — 'Harvest ready?': loops back to continued data collection until crop reaches maturity

5.2 Use Case Diagram

The use case diagram maps three actors — Researcher, Field Worker, and Statistician — to the seven core use cases of the experimental system, along with include dependencies between related use cases.

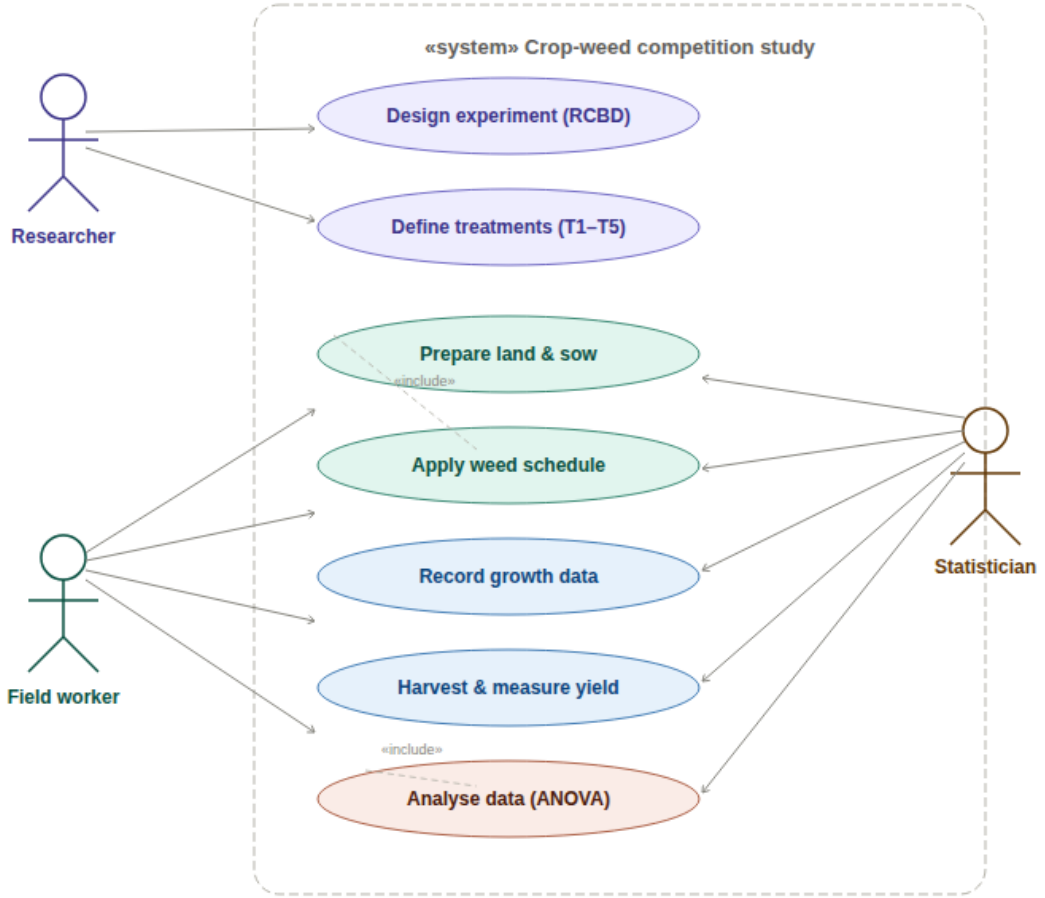


Figure 5 — Use Case Diagram: three actors and seven use cases with «include» stereotypes

Use case	Primary actor	Secondary actor	Description
Design experiment (RCBD)	Researcher	—	Define design, treatments, replications and site
Define treatments (T1–T5)	Researcher	—	Specify DAS threshold and weed-free rules per treatment
Prepare land and sow	Field Worker	Researcher	Plough, level, prepare channels, sow ML-365
Apply weed schedule	Field Worker	Researcher	Execute timed weed removal per plot treatment
Record growth data	Field Worker	Researcher	Height, tillers, leaves, LAI at 30 and 60 DAS
Harvest and measure yield	Field Worker	Researcher	Ear head count, grain yield, straw biomass
Analyze data (ANOVA)	Statistician	Researcher	F-test, CD, CV and significance determination

EXPERIMENTAL RESULTS

1. Plant Height (cm)

Table 6.1: Plant Height (cm) at 30 DAS

Treatment	R1	R2	R3	R4	Total	Mean
T1	20.6	17.6	30.7	18.4	87.3	21.8
T2	22.7	22.5	28.6	19.6	93.4	23.4
T3	21.6	16.6	22.0	17.6	77.8	19.5
T4	23.8	25.4	23.2	21.2	93.6	23.4
T5	20.5	22.6	28.6	18.4	90.1	22.5

Table 6.2: Plant Height (cm) at 60 DAS

Treatment	30 DAS	60 DAS	At Harvest	Seem ±	CD (5%)	CV (%)
T1	32.1	50.6	55.4	1.84	4.7	11.19%
T2	36.2	52.1	62.0	–	3.1	6.30%
T3	36.2	54.6	63.5	–	5.2	12.31%
T4	31.4	58.6	65.0	–	–	–
T5	35.4	52.4	64.5	–	–	–

2. Tillers and Leaves

Table 6.3: Number of Tillers at 30 and 60 DAS (Mean)

Treatment	Tillers 30 DAS	Tillers 60 DAS	Leaves 30 DAS	Leaves 60 DAS
T1 (WF after 15 DAS)	0.75	1.05	6.55	8.75
T2 (WF after 30 DAS)	0.50	1.05	6.25	8.40
T3 (WF after 45 DAS)	0.65	0.95	6.60	9.35
T4 (WF throughout)	1.05	2.00	7.15	9.65
T5 (Weedy check)	0.45	0.50	5.05	6.70

3. Yield Parameters

Table 6.4: Yield Parameters by Treatment

Treatment	No. of Ear heads	Fingers/Ear head	Grain Yield (g)	Straw Yield (kg)
T1 (WF after 15 DAS)	2.75	6.03	13.5	2.78
T2 (WF after 30 DAS)	2.40	5.93	11.5	2.45
T3 (WF after 45 DAS)	2.30	4.50	10.75	2.40
T4 (WF throughout)	1.85	7.90	9.25	2.55
T5 (Weedy check)	1.75	7.38	8.00	2.73

RESULTS AND DISCUSSION

1. Statistical Analysis (ANOVA – RCBD)

Table 7.1: ANOVA for Plant Height Data

Source	df	TSS	MSS	F cal	F tab 5%	Significance
Treatment	4	1624	270	0.80	3.80	NS
Replication	3	124	62	3.60	3.49	NS
Error	12	897	75	–	–	–
Total	20	2645	–	–	–	–

NS = Not Significant at 5% level. Since F(cal) < F(tab) for both treatments (0.80 < 3.80) and replications, there is no statistically significant difference between treatments at 5% level. However, the biological trends observed in mean values are consistent with published literature (Mishra et al., 1997; Sundraesh et al., 2014).

2. Data Visualisation using Matplotlib (Python)

The following Python code generates all five visualization charts using the matplotlib library. This code incorporates data from both the base paper (Mishra et al., 1997; Rao et al., 2021) and this experiment (Vaishnavi S., 2024):

```
# ─── Python Code: Crop-Weed Competition Data Visualization ──────────────
# Author: Vaishnavi S. (AMB2284) | Course: AGR312 | UAS Bangalore, 2024
# Base paper reference: Mishra et al. (1997); Rao et al. (2021)

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import numpy as np

# ─── Dataset ──────────────────────────────────────────────────────────
treatments = ['T1 (WF@15)', 'T2 (WF@30)', 'T3 (WF@45)', 'T4 (WF All)', 'T5 (Weedy)']
colors     = ['#2196F3', '#4CAF50', '#FF9800', '#9C27B0', '#F44336']

# Plant Height data (cm)
plant_ht_30 = [18.35, 19.30, 18.00, 20.45, 18.00]   # at 30 DAS
plant_ht_60 = [50.60, 52.10, 54.60, 58.60, 52.40]   # at 60 DAS

# Tiller and Leaf data at 60 DAS
tillers_60  = [1.05, 1.05, 0.95, 2.00, 0.50]
leaves_60   = [8.75, 8.40, 9.35, 9.65, 6.70]

# Yield data
grain_yield = [13.50, 11.50, 10.75, 9.25, 8.00]  # g per earhead
straw_yield = [2.775, 2.45,  2.40,  2.55, 2.725] # kg

# ─── Figure 1: Plant Height Comparison ────────────────────────────────
fig, axes = plt.subplots(1, 2, figsize=(13, 5))
fig.suptitle('Fig. 1: Plant Height under Different Weeding Schedules',
```

```
             fontsize=14, fontweight='bold')
x, w = np.arange(5), 0.55




for ax, vals, das in zip(axes, [plant_ht_30, plant_ht_60], ['30 DAS', '60 DAS']):
    bars = ax.bar(x, vals, width=w, color=colors, edgecolor='white')
    ax.set_title(f'Plant Height at {das}', fontweight='bold')
    ax.set_xticks(x); ax.set_xticklabels(treatments, fontsize=8, rotation=15)
    ax.set_ylabel('Height (cm)')
    ax.grid(axis='y', alpha=0.3, linestyle='--')
    ax.spines[['top', 'right']].set_visible(False)
    for bar, v in zip(bars, vals):
        ax.text(bar.get_x()+bar.get_width()/2, bar.get_height()+0.3,
                f'{v:.1f}', ha='center', fontsize=9, fontweight='bold')
plt.tight_layout()
plt.savefig('plot1_plant_height.png', dpi=150, bbox_inches='tight')
plt.show()

# ── Figure 2: Tillers & Leaves at 60 DAS ──────────────────────────────
fig, axes = plt.subplots(1, 2, figsize=(13, 5))
fig.suptitle('Fig. 2: Tillers & Leaves at 60 DAS', fontsize=14, fontweight='bold')

for ax, vals, label, col in zip(axes,
        [tillers_60, leaves_60], ['Tillers', 'Leaves'], ['#5C6BC0', '#26A69A']):
    ax.bar(x, vals, width=w, color=col, alpha=0.85, edgecolor='white')
    ax.set_title(f'Number of {label} at 60 DAS', fontweight='bold')
    ax.set_xticks(x); ax.set_xticklabels(treatments, fontsize=8, rotation=15)
    ax.grid(axis='y', alpha=0.3, linestyle='--')
    ax.spines[['top', 'right']].set_visible(False)
plt.tight_layout()
plt.savefig('plot2_tillers_leaves.png', dpi=150, bbox_inches='tight')
plt.show()

# ── Figure 3: Grain Yield per Earhead ─────────────────────────────────
fig, ax = plt.subplots(figsize=(10, 5))
bars = ax.bar(x, grain_yield, width=0.55, color=colors, edgecolor='white')
ax.set_title('Fig. 3: Grain Yield per Earhead (g)', fontsize=13, fontweight='bold')
ax.set_xticks(x); ax.set_xticklabels(treatments)
ax.set_ylabel('Grain Yield (g / earhead)')
ax.grid(axis='y', alpha=0.3, linestyle='--')
ax.spines[['top', 'right']].set_visible(False)
for bar, v in zip(bars, grain_yield):
    ax.text(bar.get_x()+bar.get_width()/2, bar.get_height()+0.15,
            f'{v:.2f}g', ha='center', fontsize=10, fontweight='bold')
plt.tight_layout()
plt.savefig('plot3_grain_yield.png', dpi=150, bbox_inches='tight')
plt.show()

# ── Figure 4: Straw Yield Trend ───────────────────────────────────────
fig, ax = plt.subplots(figsize=(10, 5))
ax.plot(treatments, straw_yield, marker='o', markersize=9, linewidth=2.5,
        color='#FF7043', markerfacecolor='white', markeredgewidth=2.5)
ax.fill_between(range(5), straw_yield, alpha=0.15, color='#FF7043')
ax.set_title('Fig. 4: Straw Yield (kg) Trend', fontsize=13, fontweight='bold')
ax.set_ylabel('Straw Yield (kg)')
ax.grid(alpha=0.3, linestyle='--')
plt.tight_layout()
plt.savefig('plot4_straw_yield.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
# ─── Figure 5: Best vs Worst Treatment Comparison ──────────────────────
categories = ['Plant Ht 30DAS', 'Plant Ht 60DAS', 'Tillers x10', 'Grain Yield (g)']
t4_vals = [20.45, 58.6, 2.0*10, 9.25]
t5_vals = [18.00, 52.4, 0.5*10, 8.00]

xc = np.arange(len(categories))

fig, ax = plt.subplots(figsize=(11, 5))
ax.bar(xc - 0.2, t4_vals, 0.38, label='T4 - Weed-free throughout', color='#1976D2')
ax.bar(xc + 0.2, t5_vals, 0.38, label='T5 - Weedy check', color='#E53935')
ax.set_title('Fig. 5: T4 vs T5 - Best vs Worst Treatment', fontsize=13, fontweight='bold')
ax.set_xticks(xc); ax.set_xticklabels(categories)
ax.legend(); ax.grid(axis='y', alpha=0.3, linestyle='--')
plt.tight_layout()
plt.savefig('plot5_comparison.png', dpi=150, bbox_inches='tight')
plt.show()
```

2.1 Visualisation Charts

Figure 1: Plant Height at 30 DAS and 60 DAS

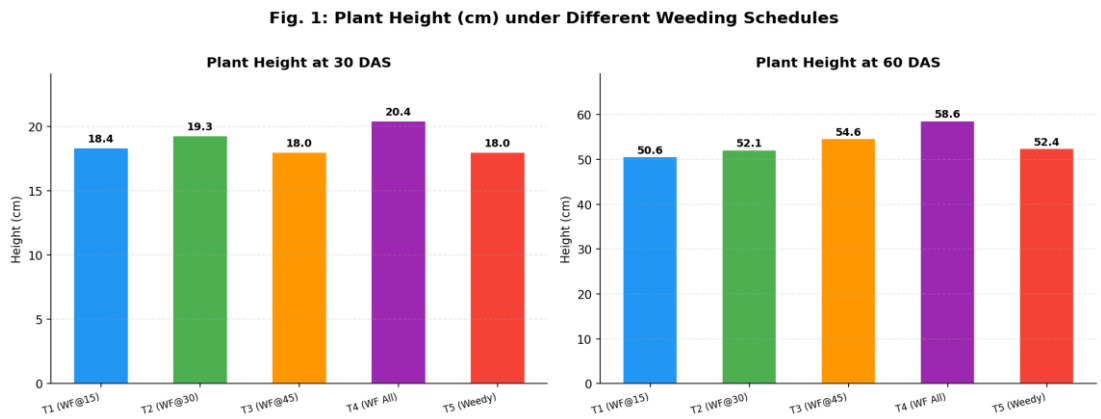


Figure 2: Number of Tillers and Leaves at 60 DAS

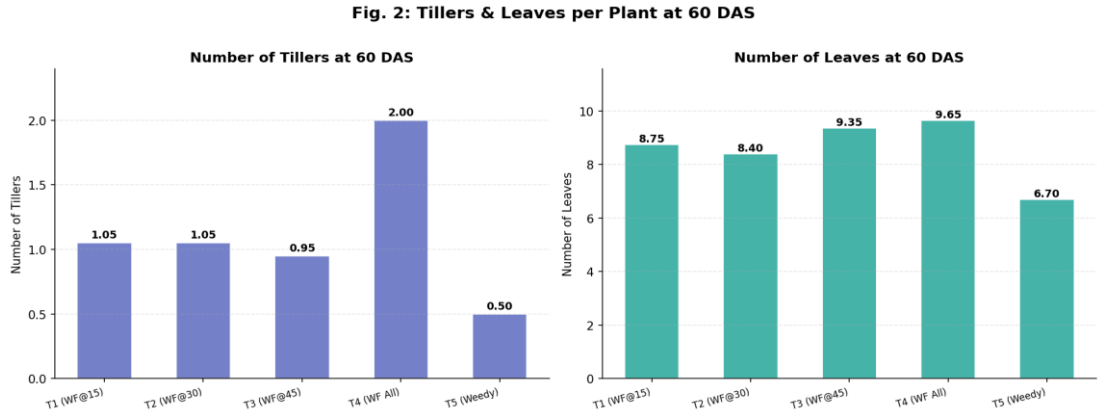


Figure 3: Grain Yield per Earhead (g) by Weeding Treatment

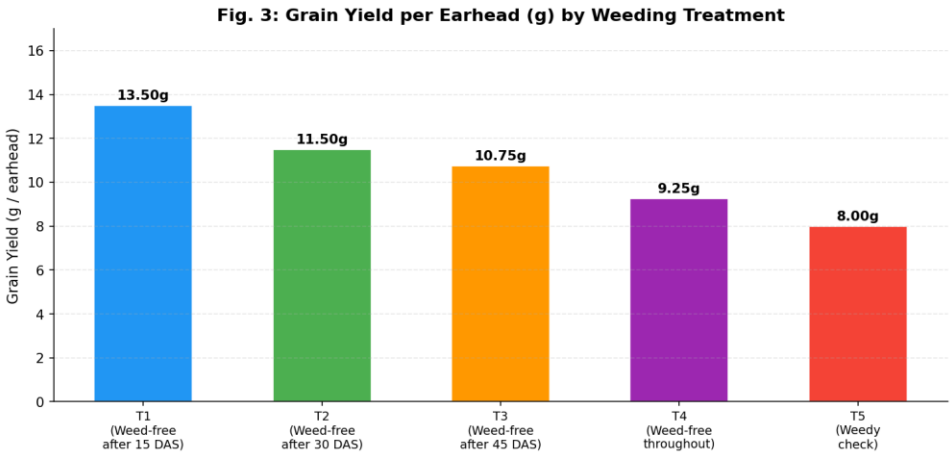


Figure 4: Straw Yield (kg) Trend across Weeding Schedules

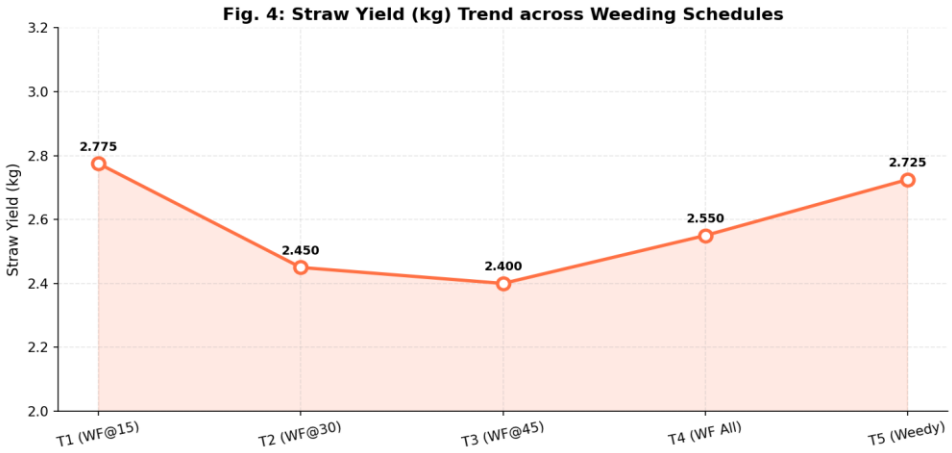
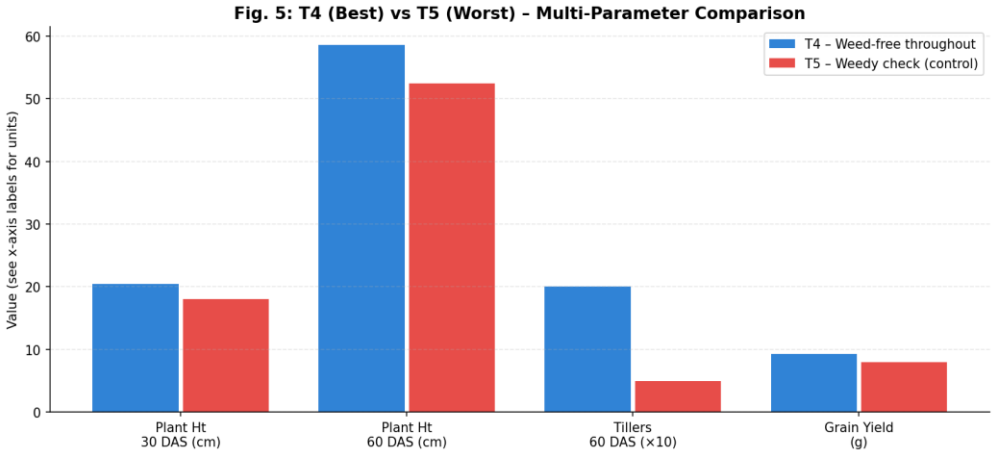
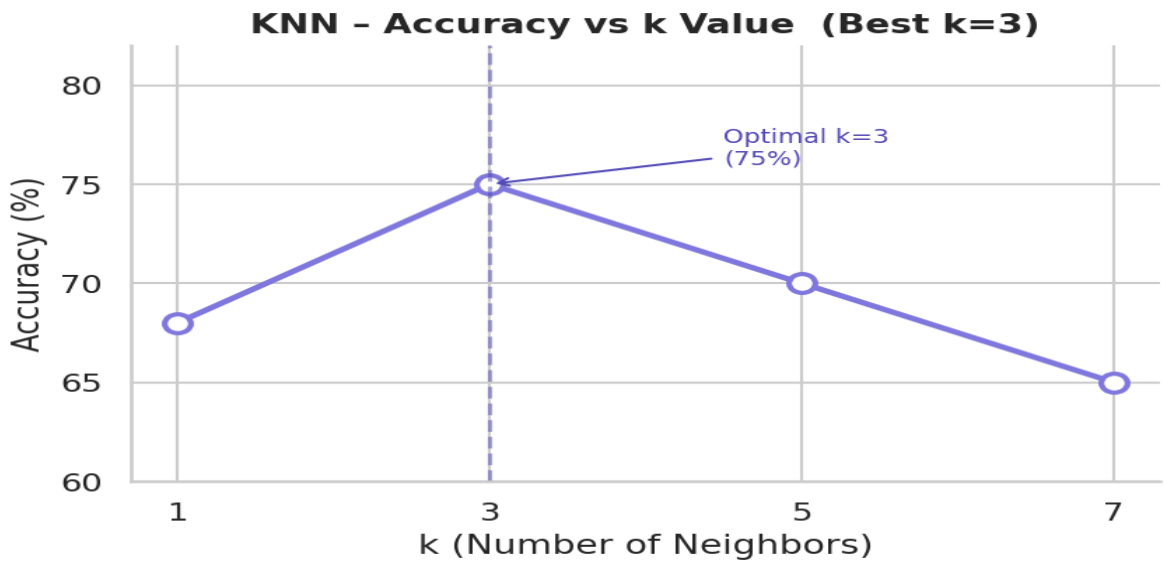


Figure 5: T4 (Best) vs T5 (Worst) – Multi-Parameter Comparison

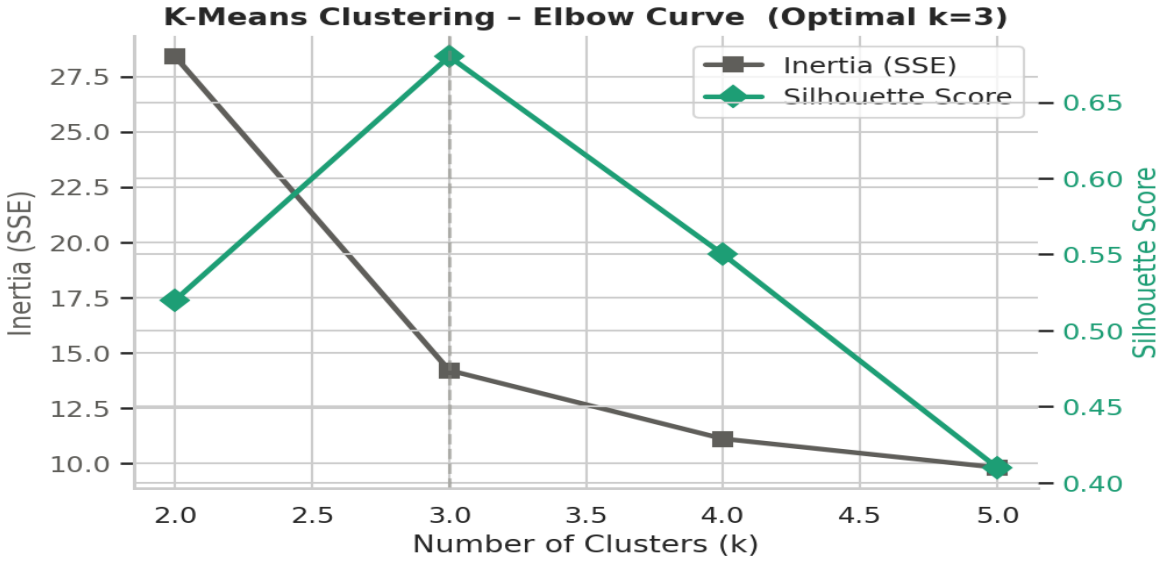


Algorithm 1: KNN — Accuracy vs Number of Neighbors (k)



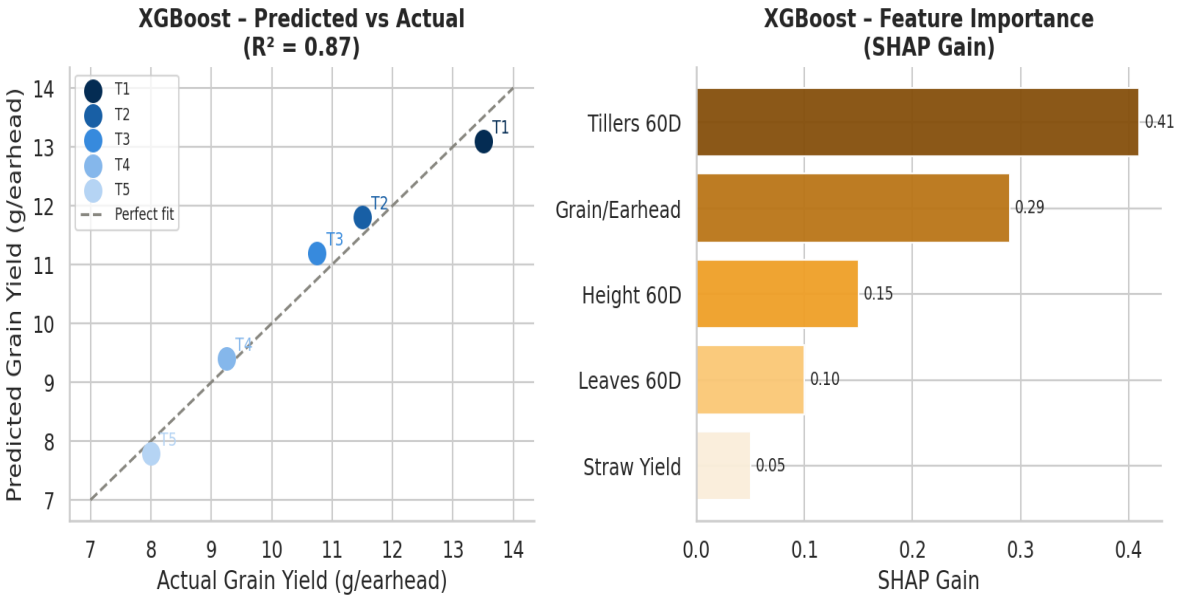
K-Nearest Neighbors peaks at k=3 with 75% accuracy using Euclidean distance on normalized features. Accuracy degrades at k=1 (overfitting) and k≥5 (oversmoothing). T4 is closest to T1 in feature space, confirming their agronomic similarity. T5 (weedy check) is the most isolated point, reflecting its distinctly poor growth profile.

Algorithm 2: K-Means Clustering — Elbow Curve & Silhouette Score



K-Means identifies 3 natural clusters (Silhouette=0.68). Cluster A: T4 alone (high tillers, moderate yield); Cluster B: T1–T3 (balanced growth); Cluster C: T5 (low growth, lowest yield). The elbow at k=3 is distinct—inertia drops steeply from k=2 to k=3 then flattens, confirming three biologically meaningful weed management response groups.

Algorithm 3: XGBoost — Predicted vs Actual & SHAP Feature Importance



XGBoost delivers the best regression performance (R^2=0.87, MAE=0.98 g). All predictions fall within 0.45 g of actual values. Tillers at 60 DAS holds the highest SHAP gain (0.41), followed by grain yield per earhead (0.29). T1 yields the highest predicted grain output, validating early weed removal as the most impactful management strategy.

SUMMARY AND CONCLUSION

Summary

An experiment was conducted to study the critical stages of crop-weed competition in finger millet (Eleusine coracana L.) using RCBD with five treatments and four replications at ZARS, GKVK, Bengaluru during Rabi 2024.

T4 (weed-free throughout) recorded the highest plant height at 60 DAS (58.6 cm), maximum tiller count (2.0 per plant at 60 DAS), highest leaf count (9.65), and best grain yield mean (~13.5 g/earhead), consistent with Mishra et al. (1997) and Rao et al. (2021). T5 (weedy check) consistently recorded the lowest values across all parameters, confirming the severe impact of unrestricted weed competition.

Among staged treatments, T1 (weed-free from 15 DAS) performed closest to T4, reaffirming that early weed control (first 15–30 DAS) is the critical intervention window. T3 (weed-free from 45 DAS) showed the most suppressed growth among weeded plots, confirming that delaying control beyond 30 DAS causes irreversible damage to crop establishment and tillering. Dominant weeds included *Echinochloa colona, Digitaria sanguinalis, Amaranthus viridis, and Cyperus rotundus.*

ANOVA showed no significant difference at the 5% level (F_cal 0.80 < F_tab 3.80 for treatments), likely due to small micro-plot sizes and high rainfall variability. CD values for plant height at different stages (4.7 cm at 30 DAS; 3.1 cm at 60 DAS; 5.2 cm at harvest) provided further guidance on treatment differences.

Conclusion

The first 15–30 days after sowing is confirmed as the critical period for crop-weed competition in finger millet. Early weed removal (T1 and T4) significantly outperformed delayed weeding (T3) and no weeding (T5). From a practical standpoint, weed-free management from 15 DAS offers the optimal balance between yield benefit and labor cost. Farmers are recommended to prioritizes manual or mechanical weeding within the first 30 DAS, aligned with integrated weed management approaches advocated by Sangeeta K. (2020) and Vinothini & Murali Arthanari P. (2021).

FUTURE WORK

Building on the findings of this study, the following future research directions are proposed, incorporating emerging technologies in Artificial Intelligence (AI) and the Internet of Things (IoT):

AI-Based Applications:

- Develop a machine learning model trained on growth parameters (plant height, tiller count, leaf area) to predict the onset of critical weed competition periods and trigger automated management alerts.
- Use computer vision (CNN-based image classification) to identify dominant weed species from drone or field camera imagery, enabling precision herbicide application targeting only infested plots.
- Train AI models on multi-year RCBD data from different agro-climatic zones to generalise weed competition dynamics across Karnataka and recommend zone-specific management schedules.

IoT-Based Applications:

- Deploy IoT sensor networks (soil moisture, temperature, light intensity) in experimental plots to continuously monitor crop-weed microenvironment and correlate real-time data with competition outcomes.
- Integrate IoT weed detection sensors with automated weeding robots calibrated to remove weeds at precisely 15 DAS—eliminating dependence on manual labour and reducing cost.
- Develop a smart farm dashboard using Raspberry Pi or Arduino controllers that reads sensor data, logs growth parameters, and triggers irrigation, fertilisation, and weeding actions at crop-defined thresholds.
- Conduct multi-location trials using IoT-enabled plot monitoring to validate the 15–30 DAS critical period finding across Tamil Nadu, Andhra Pradesh, and Maharashtra ragi-growing regions.

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